

Problem

Are there distinct prime numbers a, b, c satisfying

$$a \mid bc + b + c,$$

$$b \mid ca + c + a,$$

$$c \mid ab + a + b?$$

Source: APMO

§1 Revealing the Hidden Symmetry

Observe that

$$bc + b + c = (b + 1)(c + 1) - 1.$$

Hence

$$a \mid (b + 1)(c + 1) - 1.$$

Multiplying the right-hand side by $(a + 1)$ preserves divisibility by a :

$$a \mid (a + 1)(b + 1)(c + 1) - (a + 1).$$

Since $a \mid a$, we may add a to the expression and obtain

$$a \mid (a + 1)(b + 1)(c + 1) - 1.$$

By symmetry,

$$b \mid (a + 1)(b + 1)(c + 1) - 1,$$

and

$$c \mid (a + 1)(b + 1)(c + 1) - 1.$$

Since a, b, c are distinct primes, they are pairwise coprime. Therefore

$$\text{lcm}(a, b, c) = abc.$$

It follows that

$$abc \mid (a + 1)(b + 1)(c + 1) - 1.$$

Expanding,

$$(a + 1)(b + 1)(c + 1) - 1 = abc + ab + bc + ca + a + b + c.$$

Hence

$$abc \mid ab + bc + ca + a + b + c.$$

§2 A Size Argument

Dividing by abc gives

$$1 \leq \frac{ab + bc + ca + a + b + c}{abc} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca}.$$

Without loss of generality, assume

$$a < b < c.$$

Since a, b, c are primes,

$$b \geq a, \quad c \geq a,$$

and therefore

$$\frac{1}{ab}, \frac{1}{bc}, \frac{1}{ca} \leq \frac{1}{a^2}, \quad \frac{1}{a}, \frac{1}{b}, \frac{1}{c} \leq \frac{1}{a}.$$

Thus

$$1 \leq \frac{3}{a^2} + \frac{3}{a}.$$

If $a \geq 5$, then

$$\frac{3}{a^2} + \frac{3}{a} \leq \frac{3}{25} + \frac{3}{5} = \frac{18}{25} < 1,$$

a contradiction.

Hence

$$a \leq 3.$$

Since a is prime,

$$a \in \{2, 3\}.$$

§3 Case 1: The Smallest Prime is Two

Suppose

$$a = 2.$$

Then

$$1 \leq \frac{1}{2b} + \frac{1}{2c} + \frac{1}{bc} + \frac{1}{2} + \frac{1}{b} + \frac{1}{c}.$$

Subtracting $\frac{1}{2}$,

$$\frac{1}{2} \leq \frac{3(b+c) + 2}{2bc}.$$

Multiplying by $2bc$ gives

$$bc \leq 3(b+c) + 2.$$

Rearranging,

$$(b-3)(c-3) \leq 11.$$

Since $2 < a < b < c$ and b, c are primes, the only possibilities are

$$(b, c) = (3, 5), (3, 7), (3, 11), (5, 7).$$

We now test these candidates against

$$abc \mid ab + bc + ca + a + b + c.$$

For $(a, b, c) = (2, 3, 5)$,

$$ab + bc + ca + a + b + c = 6 + 15 + 10 + 2 + 3 + 5 = 41,$$

which is not divisible by 30.

For $(2, 3, 7)$,

$$6 + 21 + 14 + 2 + 3 + 7 = 53,$$

which is not divisible by 42.

For $(2, 3, 11)$,

$$6 + 33 + 22 + 2 + 3 + 11 = 77,$$

which is not divisible by 66.

For $(2, 5, 7)$,

$$10 + 35 + 14 + 2 + 5 + 7 = 73,$$

which is not divisible by 70.

Thus no solution exists when $a = 2$.

§4 Case 2: The Smallest Prime is Three

Suppose

$$a = 3.$$

Then $b, c \geq 5$, and

$$1 \leq \frac{1}{3} + \frac{1}{5} + \frac{1}{5} + \frac{1}{15} + \frac{1}{15} + \frac{1}{25} = \frac{68}{75} < 1,$$

a contradiction.

Therefore this case is impossible.

§5 The Final Contradiction

Both possible values of the smallest prime lead to contradictions.

Hence there do not exist distinct primes a, b, c satisfying

$$a \mid bc + b + c, \quad b \mid ca + c + a, \quad c \mid ab + a + b.$$

Therefore

No such distinct prime numbers a, b, c exist.